Notes: Saretto & Tookes (RFS, 2013)

Corporate Leverage, Debt Maturity, and Credit Supply: The Role of Credit Default Swaps

Contents

1 Big picture

- **Question.** Does the existence of traded credit default swaps (CDSs) on a firm's debt change its capital structure? The paper asks whether suppliers of debt capital lend more, and lend longer, when they can hedge issuer-specific credit risk.
- **Setting.** The sample is a panel of nonfinancial S&P 500 firms from 2002 to 2010. The paper combines Compustat, Bloomberg CDS quotes, Capital IQ capital-structure detail, Dealscan syndicated-loan data, FISD bond-underwriter data, bank Call Reports, and crisis-period write-down data.
- **Core challenge.** Firms with CDS trading on their debt are not randomly assigned. They are bigger, more visible, and more likely to use public debt. A simple CDS-versus-non-CDS comparison could therefore confuse supply-side hedging effects with selection, firm quality, or omitted demand-side determinants of leverage and maturity.
- **Main conceptual contribution.** The paper treats CDS availability as a shock to the *supply side* of credit. That shifts attention away from the usual price-of-debt margin and toward quantities and nonprice contract terms, especially debt maturity.
- **Mechanism.** CDSs can relax supply frictions through several channels:
  - banks and insurers can hedge issuer-specific credit risk and economize on regulatory capital;
  - lenders can preserve client relationships while reducing portfolio concentration risk;
  - bond investors may find corporate debt more attractive when they have a liquid hedging or resale option.
- **Main empirical design.** The paper estimates panel regressions of leverage and debt maturity on two CDS indicators: *CDS Traded* (ever has CDS during 2002–2010) and *CDS Trading* (has CDS in year t). Identification comes from the timing of CDS introduction within firms, combined with year and industry or firm fixed effects.
- **Main baseline result.** After CDS trading begins, firms maintain materially higher leverage and longer debt maturity. In the benchmark two-equation results, CDS introduction raises book leverage by about 0.050, market leverage by about 0.031, and debt maturity by roughly 1.1 to 1.8 years.
- **Key mechanism result.** The increase in debt comes mainly through *bonds*, not through bank loans. That is important because it links the results more directly to hedging by bond-market capital suppliers rather than to a generic relationship-lending story.
- **Key heterogeneity result.** The effect is strongest when hedging should matter most: for CDS firms with more liquid CDS contracts, during periods of tighter aggregate credit supply, after local same-state defaults, and when connected banks suffer larger crisis-period writedowns.
- **Key credibility result.** The main result survives: firm fixed effects, within-CDS liquidity tests, an instrumental-variables strategy for CDS introduction, matched-sample difference-in-differences around the CDS event, and placebo-style checks showing no relation between CDS introduction and future firm performance.

- **Economic takeaway.** CDS markets do not merely reprice credit risk. They expand debt capacity and lengthen debt maturity by relaxing supply constraints faced by lenders and bond investors.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper focuses on large nonfinancial firms where both bond financing and CDS trading are economically important.
- The period 2002–2010 covers the growth of single-name CDS markets and the 2007–2009 credit crisis, when the value of hedging should be especially visible.
- Finland-style household data are not available here, but the U.S. setting is rich in security-level debt data, lender relationships, underwriter links, and CDS quotes.
- Debt maturity is central because Ashcraft and Santos (2009) found only small CDS effects on spreads. If lenders respond mainly through quantity and contract terms rather than price, maturity is where the action should be.

Interpretation. The paper is built to test a supply-side hedging story exactly where that story should be strongest: large public-debt issuers in a period of important credit-market stress.

2.2 Main data sources and sample construction

- **Compustat:** firm accounting data and baseline capital-structure controls.
- **Bloomberg CDS quotes:** identifies whether a firm has actively quoted single-name CDS contracts and provides quote counts and bid-ask spreads.
- **Capital IQ:** detailed debt decomposition with principal and maturity by instrument.
- **Dealscan:** lead syndicated-loan arrangers.
- **FISD:** bond underwriters and public bond information.
- **Federal Reserve Call Reports:** bank derivatives positions used to build the instrument.
- **Bloomberg / Moody's default and writedown information:** used for local and bank-specific supply shocks.

The sample starts with nonfinancial S&P 500 firms from 2002 to 2010. About 84% are industrial firms, about 85% of those have Capital IQ debt detail, and zero-debt firm-years are excluded because maturity is undefined for them. The final benchmark sample contains **3,168 firm-year observations**, of which **1,578** have CDS trading during the year.

2.3 Leverage measures

The paper uses both book leverage and market leverage.

$$\text{Book Leverage}_{it} = \frac{\text{DLTT}_{it} + \text{DLC}_{it}}{\text{Assets}_{it}}. \tag{1}$$

$$\text{Market Leverage}_{it} = \frac{\text{Total Debt}_{it}}{\text{Firm Value}_{it}}, \tag{2}$$

where firm value is defined as book assets minus book common equity plus market equity plus deferred taxes.

Interpretation. Using both measures guards against the concern that CDS firms simply have unusual equity-price dynamics. If results line up in book and market leverage, they are less likely to be driven by denominator effects.

2.4 Debt maturity and debt decomposition

Debt maturity is measured from Capital IQ as the principal-weighted maturity of all debt components:

$$\text{Debt Maturity}_{it} = \sum_{d \in D_{it}} \omega_{dit} \cdot T_{dit}, \qquad \omega_{dit} = \frac{\text{Principal}_{dit}}{\sum_{k \in D_{it}} \text{Principal}_{kit}}. \tag{3}$$

The debt components include bonds and notes, leases, commercial paper, term loans, revolvers, trust preferred, and other obligations.

Measurement details that matter:

- if maturity is missing for a debt component, it is imputed from the firm's other debt of the same type when possible;
- if Capital IQ reports a maturity range shorter than 10 years, the midpoint is used;
- commercial-paper maturity is set to 30 days rather than the maturity of the commercial-paper program.

Interpretation. The paper is unusually strong on the maturity measure. It uses actual instrument-level maturity instead of the very coarse Compustat buckets.

2.5 CDS variables, controls, and the instrument

The two main CDS variables are:

- **CDS Trading$_{it}$:** equals 1 if the firm has quoted CDS contracts on its debt in year t.
- **CDS Traded$_i$:** equals 1 if the firm has CDS trading at any time during 2002–2010.

The core controls include market-to-book, fixed assets, profitability, size, volatility, abnormal earnings, tax credit, loss carryforward, a bond-rating dummy, an investment-grade dummy, and a commercial-paper-program dummy. The simultaneous-equation system additionally uses:

- **Industry Leverage** in the leverage equation,
- **Industry Asset Maturity** in the maturity equation.

For the IV design, the paper constructs **Bank Foreign Exchange Derivatives**: the average foreign-exchange derivatives used for hedging, not trading, relative to total assets, for the firm's lead arrangers and bond underwriters over the past five years.

2.6 Descriptive facts

The summary statistics are already informative:

- mean book leverage is **0.25**; mean market leverage is **0.16**;
- mean debt maturity is **8.68 years** (median 7.06);
- the average firm has a bond rating in 91% of firm-years and is investment grade in 72% of firm-years.

Debt is mostly bond debt:

- bonds and notes are about **73%** of total debt;
- loans and revolvers are about **16%**;
- commercial paper is about **6%**.

CDS firms differ from non-CDS firms in intuitive ways:

- book leverage: **0.27** for CDS firms versus **0.22** for non-CDS firms;
- market leverage: **0.18** versus **0.13**;
- debt maturity: **9.75** years versus **7.61** years;
- size: **9.54** versus **8.48** in log sales;
- bond share of debt: **0.77** versus **0.69**;
- loan share of debt: **0.11** versus **0.21**.

Importantly, the median rating is the same for both groups (roughly BBB), so CDS trading is not just mechanically selecting the riskiest firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's economic logic is simple:

- lenders and bond investors care about credit risk and may face capital or portfolio constraints;
- if they can hedge firm-specific credit risk with CDSs, they are more willing to supply debt;
- more elastic credit supply should show up not only in lower spreads, but also in higher leverage and longer maturity.

This generates three main hypotheses:

1. firms with traded CDS contracts should maintain more leverage;
2. firms with traded CDS contracts should maintain longer debt maturity;
3. these effects should be stronger when credit supply is tighter or when the relevant CDS contract is more liquid.

3.2 Baseline leverage equation

A stylized version of the leverage regression is:

$$\text{Leverage}_{it} = \alpha + \beta_1 \text{CDS Trading}_{it} + \beta_2 \text{CDS Traded}_i + \beta_3 \text{Debt Maturity}_{it} + \beta_4 \text{Industry Leverage}_{it} + X_{it}' \Gamma + \eta_t + \phi_{ind(i)} + \tag{4}$$

Interpretation.

- β_1 is the key coefficient: how leverage changes once CDS trading starts.
- β_2 absorbs permanent differences between firms that ever get CDSs and those that never do.
- η_t and $\phi_{ind(i)}$ remove aggregate time shocks and broad industry heterogeneity.

3.3 Baseline debt-maturity equation

The debt-maturity regression is:

$$\text{Debt Maturity}_{it} = \alpha + \delta_1 \text{CDS Trading}_{it} + \delta_2 \text{CDS Traded}_i + \delta_3 \text{Leverage}_{it} + \delta_4 \text{Industry Asset Maturity}_{it} + Z_{it}' \Theta + \eta_t + \phi_i \tag{5}$$

Interpretation. The maturity equation asks whether the availability of hedging changes a key nonprice term of borrowing. Longer maturity is interpreted as looser supply because it reduces rollover risk for the borrower and risk control for the lender.

3.4 Simultaneous-equation logic and estimation

Because leverage and maturity can be jointly chosen, the paper estimates both:

- **one-equation OLS** regressions, and
- **two-equation 2SLS** regressions treating leverage and maturity as jointly endogenous.

Identification in the simultaneous system comes from exclusion restrictions:

- **Industry Leverage** appears in the leverage equation but not the maturity equation;
- **Industry Asset Maturity** appears in the maturity equation but not the leverage equation.

Interpretation. The paper follows Johnson (2003): leverage can affect maturity through rollover risk, but peers' leverage should not directly determine a firm's maturity once its own characteristics are controlled for.

3.5 Fixed effects, standard errors, and alternative panels

The benchmark regressions include:

- year fixed effects,
- Fama-French 17-industry fixed effects,
- firm-clustered standard errors.

As a tougher specification, the paper replaces industry fixed effects and *CDS Traded* with firm fixed effects.

Interpretation. The firm-fixed-effects results answer the clean question: when the *same* firm moves from no CDS trading to CDS trading, does its leverage or maturity rise?

3.6 Liquidity specification within CDS firms

To avoid selection between CDS and non-CDS firms altogether, the paper restricts the sample to firm-years with CDS trading and replaces the binary CDS variables with liquidity measures:

$$Y_{it} = \alpha + \beta_1 \log(\text{CDS Quotes}_{it}) + X_{it}'\Gamma + \eta_t + \phi_{ind(i)} + \varepsilon_{it}, \tag{6}$$

or alternatively

$$Y_{it} = \alpha + \beta_1 \text{BidAsk Spread}_{it} + X_{it}'\Gamma + \eta_t + \phi_{ind(i)} + \varepsilon_{it}. \tag{7}$$

Interpretation. More quote activity and lower bid-ask spreads mean hedging is easier and cheaper. If the mechanism is truly about hedging, more liquid CDS contracts should be associated with more leverage and longer maturity.

3.7 Instrumental-variables design for CDS timing

To address the concern that CDS markets arise exactly when a firm's leverage or maturity is about to change, the paper instruments for *CDS Trading* using banks' hedging propensity:

$$\Pr(\text{CDS Trading}_{it} = 1) = F\big(\pi_0 + \pi_1 \text{Bank FX Derivatives}_{i,t-1} + X_{it}'\Pi + \eta_t + \phi_{ind(i)}\big), \tag{8}$$

where the fitted value from this probit is used as the instrument.

Interpretation. The idea is that banks and underwriters that hedge foreign-exchange risk are more likely to use credit derivatives in general, but their FX hedging should not directly determine a U.S. firm's leverage or debt maturity.

3.8 Difference-in-differences around CDS introduction

The paper also isolates the CDS event using matched firms without CDSs. For each treated firm, a non-CDS S&P 500 firm with a similar propensity score is selected. The paper then studies changes from the end of year $t-1$ to the end of year t, and from the end of year $t-1$ to the end of year $t+1$:

$$\Delta Y_i^{(t-1,t)} = \big(Y_{i,t} - Y_{i,t-1}\big) - \big(Y_{m(i),t} - Y_{m(i),t-1}\big), \tag{9}$$

with an analogous definition for $t+1$.

Interpretation. This design asks whether leverage and maturity move differently for newly treated CDS firms relative to ex-ante similar untreated firms precisely around the introduction event.

3.9 Credit-supply tightening tests

The supply-side story predicts stronger CDS effects when credit supply tightens. The paper studies this three ways:

- year-by-year estimates and comparison with the Senior Loan Officer Opinion Survey;
- local same-state default shocks,
- bank-specific crisis writedown shocks.

A generic interaction design is:

$$Y_{it} = \alpha + \beta_1 \text{CDS Trading}_{it} + \beta_2 Shock_{it} + \beta_3 \big(Shock_{it} \times \text{CDS Trading}_{it}\big) + X_{it}'\Gamma + \eta_t + \phi_{ind(i)} + \varepsilon_{it}. \tag{10}$$

Interpretation. The interaction coefficient asks whether CDS firms are relatively protected when the local or lender-specific supply of credit comes under pressure.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the baseline estimate

The benchmark estimate is identified from the timing of CDS introduction, not from a static comparison of firms with and without CDSs.

- *CDS Traded* absorbs permanent level differences between eventual CDS firms and other firms.
- Firm fixed effects go further and use only within-firm variation in CDS status.
- Year fixed effects remove aggregate credit-market and business-cycle shocks.

- Industry fixed effects remove broad cross-industry leverage and maturity differences.

4.2 Why the interpretation is about credit supply rather than debt demand

Several features push the evidence toward a supply-side reading:

- the paper focuses on leverage and maturity, not just spreads;
- the effect is stronger when external credit conditions deteriorate;
- the debt increase occurs mainly in public bonds, where hedging and regulatory-capital relief are directly relevant;
- same-industry peers' CDS markets do not produce similar spillover effects.

Interpretation. If CDSs mainly reflected firms' own future debt demand, the strong interaction with supply shocks and the bond-versus-loan decomposition would be harder to explain.

4.3 Selection and time-invariant unobservables

A major challenge is that CDS firms are larger and more bond-market oriented. The paper addresses this three ways:

- **CDS Traded** controls for permanent CDS-firm differences;
- **firm fixed effects** difference out all time-invariant firm heterogeneity;
- **within-CDS liquidity tests** use only firms that already have CDS contracts.

These three approaches all point in the same direction.

4.4 Timing endogeneity and the IV exclusion restriction

The remaining concern is simultaneity: lenders might create CDS markets exactly when they foresee leverage or maturity changes. The IV tries to break that link.

- The identifying variable is banks' foreign-exchange derivatives used for hedging, relative to assets.
- The first stage is strong: the incremental pseudo-R^2 is about **1.4%**, and the incremental F-tests are about **52.6** and **51.7**.
- After size and having a credit rating, bank FX hedging is one of the strongest predictors of CDS trading.

Interpretation. The exclusion assumption is that banks' general hedging propensity affects firm capital structure only through the likelihood that the firm has a tradable CDS market.

4.5 Matched-sample event design and pretrend logic

The difference-in-differences design uses one nearest-neighbor propensity-score match for each treated firm.

- The final event-study sample has **122 treated firms** with matched controls.
- Matching uses lagged covariates and lagged changes in leverage and maturity to make pre-event trends more comparable.
- The average propensity-score gap between treated and control firms is about **1.3%**.

Interpretation. This design is not perfect random assignment, but it makes the question much tighter: what happens to leverage and maturity right around the CDS adoption event relative to comparable untreated firms?

4.6 Alternative explanations the paper addresses

- **Same-industry spillovers:** Appendix Table A2 shows that competitors' CDS markets do not explain the results.
- **Future fundamentals:** Appendix Table A7 finds no significant impact of *CDS Trading* on future firm performance measures.
- **Asset-side growth:** the matched-sample exercise measures debt growth directly and still finds a strong increase after CDS introduction.
- **Loan-only relationship lending story:** Appendix Table A3 shows the effect is mostly in bonds, not loans.

4.7 Limits of interpretation

- The sample is large-cap U.S. public firms, so external validity to smaller or private firms is limited.
- The instrument is plausible but not beyond debate; it depends on the exclusion from firms' leverage and maturity choices.
- The paper cannot cleanly decompose which supply channel dominates: regulatory capital relief, portfolio hedging, bond-market liquidity, or a resale option.
- CDS quote data identify contracts with meaningful market activity, but the exact start of economically relevant hedging opportunities may precede the first observed quote.

4.8 Relation to the literature

The paper sits at the intersection of three literatures:

- supply-side frictions in corporate finance,

- corporate debt maturity choice,
- and the real effects of CDS markets.

Its distinctive contribution is to show that CDS markets matter even when spread effects appear small, because quantity and maturity margins respond strongly.

5 Tables and figures

5.1 Table 1: sample structure and debt composition

Table 1
Summary statistics

	S&P 500					S&P 500 and CDS Trading					S&P 500 and no CDS Trading				
	Mean	Median	Stdev	Min	Max	Mean	Median	Stdev	Min	Max	Mean	Median	Stdev	Min	Max
Panel A: Firm variables															
Book Leverage	0.25	0.23	0.14	0.00	1.39	0.27	0.26	0.13	0.00	0.75	0.22	0.20	0.15	0.00	1.39
Market Leverage	0.16	0.13	0.12	0.00	0.71	0.18	0.16	0.11	0.00	0.56	0.13	0.10	0.12	0.00	0.71
Debt Maturity	8.68	7.06	6.60	0.08	72.45	9.75	8.41	6.75	0.08	72.45	7.61	5.89	6.27	0.08	59.53
Asset Maturity	8.09	6.17	6.05	0.45	71.82	8.54	6.81	5.83	0.45	35.15	7.65	5.59	6.23	0.46	71.82
Market to Book	1.93	1.64	0.99	0.57	13.73	1.71	1.53	0.68	0.57	7.44	2.15	1.80	1.19	0.58	13.73
Fixed Asset	0.31	0.25	0.23	0.00	0.94	0.34	0.28	0.23	0.00	0.93	0.29	0.22	0.22	0.01	0.94
Profitability	0.11	0.10	0.08	−0.33	0.86	0.10	0.09	0.07	−0.33	0.64	0.11	0.10	0.09	−0.26	0.86
Size	9.01	8.97	1.15	4.62	12.96	9.54	9.41	1.00	6.91	12.95	8.48	8.43	1.04	4.62	12.96
CP Program	0.31	0.00	0.46	0.00	1.00	0.36	0.00	0.48	0.00	1.00	0.25	0.00	0.44	0.00	1.00
Volatility	0.03	0.02	0.04	0.00	0.38	0.03	0.02	0.04	0.00	0.38	0.04	0.02	0.04	0.00	0.37
Abnormal Earnings	0.01	0.01	0.34	−9.49	7.89	0.02	0.01	0.25	−2.11	4.21	0.01	0.01	0.41	−9.49	7.89
Investment Tax Credit	0.04	0.02	0.05	−0.00	0.28	0.05	0.02	0.06	0.00	0.28	0.04	0.02	0.05	−0.00	0.26
Loss Carry Forward	0.05	0.00	0.21	0.00	5.48	0.03	0.00	0.07	0.00	1.01	0.07	0.00	0.29	0.00	5.48
Rating	8.61	9.00	3.97	0.00	27.00	9.31	9.00	2.84	0.00	27.00	7.93	9.00	4.73	0.00	27.00
Rated	0.91	1.00	0.28	0.00	1.00	1.00	1.00	0.04	0.00	1.00	0.83	1.00	0.38	0.00	1.00
Investment Grade	0.72	1.00	0.45	0.00	1.00	0.82	1.00	0.38	0.00	1.00	0.62	1.00	0.48	0.00	1.00

(*continued*)

Table 1
Continued

	S&P 500					S&P 500 and CDS Trading					S&P 500 and no CDS Trading				
	Mean	Median	Stdev	Min	Max	Mean	Median	Stdev	Min	Max	Mean	Median	Stdev	Min	Max
Panel B: Debt decomposition															
Bond/Debt	0.73	0.83	0.29	0.00	1.00	0.77	0.85	0.25	0.00	1.00	0.69	0.80	0.32	0.00	1.00
Bond Maturity	9.47	7.87	7.48	0.00	75.45	10.76	9.16	7.85	0.00	75.45	8.20	6.61	6.87	0.00	66.05
Lease/Debt	0.02	0.00	0.09	0.00	1.00	0.02	0.00	0.05	0.00	0.60	0.03	0.00	0.12	0.00	1.00
Lease Maturity	9.75	5.00	13.78	0.00	88.50	10.25	5.46	12.63	0.00	86.50	9.22	4.50	14.91	0.00	88.50
Loan/Debt	0.16	0.00	0.32	0.00	2.00	0.11	0.00	0.22	0.00	1.96	0.21	0.00	0.39	0.00	2.00
Loan Maturity	1.53	0.50	2.74	0.00	31.25	1.50	0.50	2.91	0.00	31.25	1.56	0.50	2.55	0.00	29.08
CP/Debt	0.06	0.00	0.14	0.00	1.00	0.06	0.00	0.13	0.00	1.00	0.06	0.00	0.16	0.00	1.00
CP Maturity	0.08	0.08	0.00	0.08	0.08	0.08	0.08	0.00	0.08	0.08	0.08	0.08	0.00	0.08	0.08
TP/Debt	0.01	0.00	0.06	0.00	0.88	0.01	0.00	0.07	0.00	0.84	0.01	0.00	0.05	0.00	0.88
TP Maturity	25.47	26.44	11.72	0.58	55.66	26.70	28.50	10.26	0.58	54.50	23.22	21.50	13.81	0.70	55.66
Other/Debt	0.05	0.00	0.13	0.00	1.00	0.06	0.01	0.13	0.00	0.98	0.04	0.00	0.12	0.00	1.00
Other Maturity	4.51	2.00	6.24	0.00	42.50	5.12	2.00	7.06	0.00	42.50	3.52	2.00	4.49	0.00	29.50

This table presents summary statistics for the sample of nonfinancial firms in the S&P 500 index during the 2002–2010 period. Panel A presents summary statistics for firm-level variables used in the main analyses. Panel B presents summary statistics of the debt decomposition variables. *Book Leverage* is book leverage, defined as total debt (long-term debt plus debt in current liabilities), divided by the book value of assets. *Market Leverage* is market leverage, defined as total debt, divided by firm value, where firm value is defined as the book value of assets, minus the book value of common equity, plus the market value of equity, plus the book value of deferred taxes. *Debt Maturity* is debt maturity in years, defined as the principal-weighted maturity of all debt, as reported in Capital IQ. *Asset Maturity* is the weighted maturity of the firm's assets, defined as: (gross PPE divided by depreciation expense, times gross PPE divided by total assets) plus (current assets divided by cost of goods sold, times current assets divided by total assets). *Market to Book* is the firm's market-to-book ratio, defined as the market value of assets (equity market capitalization plus the book value of other liabilities), divided by the book value of assets. *Fixed Assets* is defined as net PPE, divided by the book value of total assets. *Profitability* is defined as earnings before interest and taxes, divided by the book value of assets. Size is the natural logarithm of total sales, in millions of U.S. Dollars. *CP Program* is a dummy equal to one if the firm has a commercial paper program at the beginning of year t, *Volatility* is defined as the standard deviation of annual changes in earnings over years t through $t-5$, divided by the book value of assets at the end of year t, *Abnormal Earnings* is defined as the change in earnings per share from year $t-1$ to year t, divided by the share price as the end of year t, Tax Credit is defined as the investment tax credit, divided by the book value of assets. *Loss Carry Forward* is defined as the net operating loss carry forwards, divided by the book value of assets. *Rating* is the S&P credit rating of the firm, where the value 1 corresponds to an S&P rating of AAA; 2 corresponds to AA+, and so on. *Rated* is a dummy variable equal to one if the firm has a Standard and Poor's rating. *Investment Grade* is a dummy equal to one if the firm has an investment grade rating (i.e., BBB or higher). *Bond/Debt* is the fraction of total debt that are bonds and notes. *Bond Maturity* is the maturity of the firm's bonds and notes, in years. *Lease/Debt* is the fraction of total debt that are lease obligations. *Lease Maturity* is the maturity of lease obligations, in years. *Loan/Debt* is the fraction of total debt that are term loans or revolvers. *Loan Maturity* is the maturity of term and revolver loan obligations, in years. *CP/Debt* is the fraction of commercial paper in total debt. *CP Maturity* is set equal to 30 days. *TP/Debt* is the fraction of trust preferred obligations in total debt. *TP Maturity* is defined as the maturity of trust preferred obligations, in years. *Other/Debt* is the fraction of other obligations in total debt. *Other Maturity* is defined as the maturity of other obligations. "S&P 500 and CDS" indicates that the firm is in the S&P 500 sample and has traded CDS contracts on its debt (CDS quoted in Bloomberg) during year t, "S&P and No CDS" indicates that the firm does not have traded CDS on its debt.

Read:

- The average firm has book leverage 0.25, market leverage 0.16, and debt maturity 8.68 years.
- CDS firms are more levered and have longer maturity than non-CDS firms even in raw data: 0.27 versus 0.22 in book leverage and 9.75 versus 7.61 years in debt maturity.
- Bonds dominate the debt structure. On average, bonds are 73% of debt, loans 16%, and commercial paper 6%.
- CDS firms rely more on bonds and less on loans than non-CDS firms.

Economic message. The raw data already point to a CDS-bond market connection and motivate the later decomposition showing that CDS availability works mainly through bond financing.

5.2 Table 2: predicted signs and the system design

Read:

- Table 2 lays out the expected sign of every regressor in the leverage and maturity equations.
- The main variable of interest, *CDS Trading*, is predicted to have a positive sign in both equations.
- The table also shows the exclusion restrictions: industry leverage belongs only in the leverage equation, and industry asset maturity only in the maturity equation.

Table 2
Predicted signs

Variable	Leverage Equation Sign	Leverage Equation Explanation	Maturity Equation Sign	Maturity Equation Explanation
CDS Trading	+	CDSs allow suppliers of capital to hedge risk and increase their willingness to extend credit.	+	CDSs allow suppliers of capital to hedge risk and increase their willingness to extend longer maturity credit.
CDS Traded	+/−	This variable controls for unobservable differences between firms that eventually have CDS versus non-CDS firms.	+/−	This variable controls for unobservable differences between firms that eventually have CDS versus non-CDS firms.
Leverage			+	Firms face liquidity risk when they roll over short-term debt. Firms with high leverage choose more long maturity debt to avoid excessive liquidation.
Debt Maturity	+	Firms face liquidity risk when they roll over short-term debt. Firms with short maturity debt will therefore choose less leverage.		
Industry Asset Maturity			+	Maturity matching of assets and liabilities can reduce underinvestment problems (Myers 1977). Firms in the same industries have common characteristics that might impact asset maturities.
Industry Leverage	+	Firms in the same industries have common characteristics that might impact leverage ratios.		
Market to Book	−	Firms with high growth opportunities should use low leverage to mitigate underinvestment incentives (Myers 1977).	−	Firms with high growth opportunities will choose short maturity debt so that debt is due before options to invest expire, thereby mitigating underinvestment problems.
Fixed Assets	+	Tangible assets have high collateral values, increasing firms' debt capacity.	+	Fixed assets are included because Graham, Li, and Qiu (2008) find that asset tangibility is related to loan maturity.
Profitability	+/−	The free cash flow hypothesis (e.g., Jensen 1986) predicts that firms with high profitability will take on greater leverage since interest payments discipline wasteful spending incentives. In contrast, Myers (1984) argues that firms have a pecking order, first using internal funds. High profit firms would therefore have less leverage.	+	Profitability is included in the maturity equation due to recent empirical findings by Graham, Li, and Qiu (2008) who find that profitability is related to loan maturity.

(*continued*)

Table 2
Continued

Variable	Leverage Equation Sign	Leverage Equation Explanation	Maturity Equation Sign	Maturity Equation Explanation
Size	+/−	Larger firms are more diversified, so may have greater debt capacity (Lewellen 1971). Larger firms also have fewer asymmetric information problems so will have greater access to equity markets, decreasing debt.	+	Diamond (1991) predicts an increasing and then decreasing relationship between debt maturity and credit quality/liquidity risk.
CP Program	+	Firms with commercial paper programs are expected to have higher leverage due to access to these additional debt markets (as in Faulkender and Petersen 2006)	−	Diamond (1991) predicts an increasing and then decreasing relationship between debt maturity and credit quality/liquidity risk.
Volatility	−	Increased earnings volatility increases default probabilities, decreasing debt capacity.	−	Increased earnings volatility increases default probabilities, decreasing optimal maturity.
Abnormal Earnings	+	If there is a signaling effect associated with leverage choice (Ross 1977), leverage will be positively related to abnormal earnings.	−	If there is a signaling effect associated with maturity choice (as in Diamond 1991 and Flannery 1986), maturity will be negatively related to abnormal earnings.
Tax Credit	−	Alternative tax shields can reduce the value of long term debt.	−	Alternative tax shields can reduce the value of long-term debt (e.g., Brick and Ravid 1985) in which upward sloping yield curve makes long term-debt more valuable.
Loss Carry Forward	−	Alternative tax shields can reduce the value of long-term debt.	−	Alternative tax shields can reduce the value of long-term debt.
Rated	+	Rated firms are expected to have higher leverage due to access to public debt markets (as in Faulkender and Petersen 2006).	+	Unrated firms have lower credit quality and may find it more difficult to borrow longer-term debt.
Investment Grade	+/−	Control variable. The assets underlying CDSs tend to be investment grade. We include this dummy variable to ensure that our inferences about the impact of CDS are not due to differences in credit rating.	+/−	Control variable. The assets underlying CDSs tend to be investment grade. We include this dummy variable to ensure that our inferences about the impact of CDS are not due to differences in credit rating.

This table presents predicted signs of the coefficients in the simultaneous equations model of leverage and maturity.

Economic message. The paper is not a black-box regression exercise. The structure of the system is directly tied to standard capital-structure and debt-maturity theories, with CDS availability layered on top as the new supply-side variable.

5.3 Table 3 and Appendix Table A3: baseline CDS effects

Table 3
Corporate leverage, debt maturity, and credit default swaps

	Book Leverage				Market Leverage			
	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Leverage regression								
Debt Maturity	0.001	0.000	0.005	−0.002	0.000	−0.000	0.002	−0.003
	(1.45)	(0.67)	(0.77)	(−0.48)	(0.79)	(−0.03)	(0.43)	(−0.69)
Industry Book Leverage	0.367**	0.194**	0.373**	0.202**				
	(6.49)	(2.85)	(6.59)	(2.84)				
Industry Market Leverage					0.375**	0.273**	0.380**	0.281**
					(8.55)	(5.34)	(7.91)	(5.41)
Market to Book	−0.006	−0.013**	−0.005	−0.013**	−0.030**	−0.026**	−0.030**	−0.026**
	(−0.78)	(−2.92)	(−0.76)	(−2.92)	(−5.42)	(−4.13)	(−5.38)	(−4.13)
Fixed Asset	0.095**	0.044	0.080*	0.044	0.055**	0.000	0.048*	−0.000
	(2.93)	(0.85)	(1.88)	(0.85)	(2.56)	(0.00)	(1.65)	(−0.01)
Profitability	0.130*	−0.076	0.140*	−0.079	−0.075*	−0.167**	−0.070	−0.169**
	(1.80)	(−1.53)	(1.94)	(−1.59)	(−1.72)	(−4.47)	(−1.59)	(−4.51)
Size	−0.023**	−0.022**	−0.023**	−0.023**	−0.006	−0.010	−0.006	−0.010
	(−4.37)	(−1.99)	(−4.44)	(−2.04)	(−1.58)	(−1.22)	(−1.63)	(−1.27)
Volatility	−0.059	−0.037	−0.037	−0.053	−0.051	−0.071	−0.042	−0.084
	(−0.54)	(−0.33)	(−0.33)	(−0.45)	(−0.77)	(−1.01)	(−0.63)	(−1.18)
Abnormal Earnings	0.007	0.000	0.007	0.001	0.005	0.002	0.005	0.002
	(1.12)	(0.08)	(1.17)	(0.10)	(0.98)	(0.35)	(1.00)	(0.36)
Tax Credit	0.079	−0.103	0.015	−0.080	0.006	−0.066	−0.023	−0.046
	(0.73)	(−0.67)	(0.10)	(−0.49)	(0.08)	(−0.62)	(−0.20)	(−0.42)
Loss Carry Forward	0.027	0.018	0.024	0.015	0.011	−0.001	0.010	−0.004
	(1.40)	(0.54)	(1.20)	(0.47)	(1.09)	(−0.06)	(0.88)	(−0.15)
Rated	0.111**	0.137**	0.112**	0.140**	0.081**	0.067**	0.082**	0.069**
	(5.27)	(3.71)	(5.27)	(3.74)	(6.28)	(3.75)	(6.25)	(3.77)
Investment Grade	−0.090**	−0.032**	−0.093**	−0.032**	−0.081**	−0.034**	−0.082**	−0.034**
	(−7.14)	(−2.99)	(−6.75)	(−2.97)	(−8.49)	(−3.82)	(−7.36)	(−3.80)
CP Program	0.019**	0.005	0.027*	0.005	0.004	0.003	0.008	0.003
	(2.18)	(1.15)	(1.74)	(1.12)	(0.74)	(0.84)	(0.70)	(0.80)
CDS Traded	0.004		0.001		0.001		−0.000	
	(0.34)		(0.07)		(0.14)		(−0.04)	
CDS Trading	0.055**	0.018**	0.050**	0.020**	0.033**	0.009*	0.031**	0.011*
	(5.33)	(2.43)	(3.68)	(2.30)	(4.76)	(1.70)	(3.25)	(1.81)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X		X		X		X	
Firm Fixed Effects		X		X		X		X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.340	0.793	0.339	0.793	0.537	0.837	0.537	0.837

(*continued*)

Table 3
Continued

	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Debt maturity regression								
Book Leverage	2.017	1.184	−10.339	−3.256				
	(1.05)	(0.72)	(−1.17)	(−0.14)				
Market Leverage					0.242	0.166	−16.332	2.282
					(0.10)	(0.07)	(−1.53)	(0.15)
Industry Asset Maturity	0.239*	0.481**	0.298**	0.484**	0.248*	0.482**	0.312**	0.481**
	(1.81)	(2.71)	(2.05)	(2.74)	(1.87)	(2.70)	(2.17)	(2.73)
Market to Book	−0.004	−0.031	−0.144	−0.092	−0.019	−0.043	−0.588	0.015
	(−0.02)	(−0.13)	(−0.55)	(−0.24)	(−0.07)	(−0.18)	(−1.42)	(0.03)
Fixed Asset	1.153	−2.147	2.533	−1.897	1.363	−2.082	2.453	−2.105
	(0.62)	(−0.94)	(1.23)	(−0.71)	(0.73)	(−0.91)	(1.26)	(−0.92)
Profitability	−2.075	−0.484	−0.375	−0.842	−1.776	−0.549	−3.207	−0.163
	(−0.64)	(−0.20)	(−0.11)	(−0.27)	(−0.55)	(−0.22)	(−0.94)	(−0.04)
Size	0.105	−0.385	−0.174	−0.482	0.060	−0.409	−0.019	−0.392
	(0.37)	(−0.73)	(−0.48)	(−0.64)	(0.22)	(−0.78)	(−0.07)	(−0.72)
Volatility	−4.436	−4.994	−4.949	−5.139	−4.515	−5.023	−4.786	−4.900
	(−1.00)	(−1.14)	(−1.13)	(−1.17)	(−1.02)	(−1.16)	(−1.10)	(−1.09)
Abnormal Earnings	−0.032	0.041	0.032	0.040	−0.023	0.040	0.036	0.037
	(−0.16)	(0.27)	(0.15)	(0.27)	(−0.11)	(0.27)	(0.17)	(0.24)
Tax Credit	12.339**	6.885	13.783**	6.516	12.574*	6.795	12.596*	6.892
	(1.86)	(0.85)	(2.01)	(0.82)	(1.91)	(0.84)	(1.95)	(0.86)
Loss Carry Forward	0.926	−0.720	1.150	−0.637	0.962	−0.698	1.011	−0.697
	(1.15)	(−0.58)	(1.39)	(−0.49)	(1.21)	(−0.57)	(1.24)	(−0.57)
Rated	−0.389	0.799	1.104	1.430	−0.166	0.956	1.300	0.805
	(−0.42)	(0.74)	(0.82)	(0.43)	(−0.18)	(0.89)	(1.02)	(0.57)
Investment Grade	1.134*	0.116	−0.038	−0.031	0.964	0.082	−0.457	0.157
	(1.84)	(0.18)	(−0.04)	(−0.03)	(1.52)	(0.12)	(−0.42)	(0.19)
CP Program	−1.825**	0.015	−1.577**	0.039	−1.786**	0.021	−1.685**	0.014
	(−4.19)	(0.05)	(−3.45)	(0.13)	(−4.13)	(0.07)	(−3.91)	(0.05)
CDS Traded	0.928		1.043		0.945		1.017	
	(1.33)		(1.47)		(1.35)		(1.46)	
CDS Trading	1.091*	0.675*	1.788**	0.756	1.196**	0.695*	1.748**	0.676
	(1.90)	(1.76)	(2.44)	(1.23)	(2.10)	(1.82)	(2.67)	(1.59)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X		X		X		X	
Firm Fixed Effects		X		X		X		X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.123	0.678	0.124	0.678	0.122	0.678	0.125	0.678

Panel A presents ordinary least squares (One Equation) and two-stage least squares (Two Equations) regression results of annual leverage on the explanatory variables from Johnson (2003), plus two credit default swap variables, an industry leverage control, and debt rating dummies. Panel B reports the results of the maturity regression. *CDS Trading* is a dummy variable equal to one if the firm has quoted CDS contracts on its debt during year *t*, We also add *CDS Traded*, a dummy variable equal to one if the firm has a traded CDS during the 2002–2010 sample period. The explanatory variables not already defined in Table 1 are *Industry Leverage* and *Industry Asset Maturity*, defined as the median leverage and the median asset maturity of all firms in the 3-digit SIC code. *Industry Leverage* is based on industry book leverage in the *Book Leverage* regressions and on industry market leverage in the *Market Leverage* regressions. The regression specifications shown in Columns 1, 3, 5, and 7 include time and industry fixed effects. Industry fixed effects and the *CDS Traded* dummy variable are replaced with firm fixed effects in the specifications shown in Columns 2, 4, 6, and 8. In Columns 3, 4, 7, and 8 both leverage and maturity are treated as endogenous variables and their equations are estimated simultaneously using two stage least squares. The identifying variable for the leverage equation is *Industry Leverage*. The identifying variable for the debt maturity equation is *Industry Asset Maturity*. Results of the first-stage regressions are reported in Appendix Table A1. Standard errors are clustered at the firm level in all regressions. * and ** denote significance levels of 10% and 5%, respectively. Constants are estimated but not reported. The sample is composed of nonfinancial firms in the S&P 500 index during the 2002–2010 period granting a total of 3,168 firm/year observations.

Read:

- In the benchmark two-equation results, CDS introduction raises book leverage by about **0.050** and market leverage by about **0.031**.

- In the maturity equation, CDS trading raises debt maturity by about **1.09 to 1.79 years** depending on the exact specification.
- These results remain positive in the firm-fixed-effects versions, though somewhat smaller.
- Appendix Table A3 shows that the debt increase comes mainly through **bond debt**, not through loans.

Economic message. Once the firm's debt can be hedged, the firm keeps more debt outstanding and keeps it outstanding for longer. The bond-market channel is doing the heavy lifting.

5.4 Table 4 and Appendix Table A4: liquidity and within-CDS tests

Table 4
CDS liquidity proxies

	Book Leverage				Market Leverage			
	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Leverage regression								
Debt Maturity	−0.001	−0.001*	−0.002	−0.004	−0.001**	−0.001**	−0.004	−0.005**
	(−1.62)	(−1.75)	(−0.42)	(−1.04)	(−1.99)	(−2.05)	(−1.57)	(−1.98)
Industry Book Leverage	0.363**	0.352**	0.359**	0.330**				
	(4.55)	(4.43)	(4.28)	(3.95)				
Industry Market Leverage					0.330**	0.325**	0.303**	0.288**
					(5.29)	(5.22)	(4.58)	(4.37)
Market to Book	−0.007	−0.003	−0.007	−0.002	−0.057**	−0.055**	−0.057**	−0.055**
	(−0.67)	(−0.25)	(−0.66)	(−0.23)	(−8.20)	(−7.79)	(−8.17)	(−7.72)
Fixed Asset	0.094**	0.097**	0.097**	0.110**	0.052**	0.053**	0.066**	0.072**
	(2.46)	(2.53)	(2.23)	(2.53)	(1.98)	(2.03)	(2.16)	(2.38)
Profitability	0.092	0.080	0.091	0.077	−0.032	−0.038	−0.035	−0.044
	(0.95)	(0.84)	(0.94)	(0.81)	(−0.49)	(−0.58)	(−0.53)	(−0.67)
Size	−0.023**	−0.021**	−0.023**	−0.020**	−0.010*	−0.009*	−0.009*	−0.007
	(−3.11)	(−2.93)	(−3.07)	(−2.67)	(−1.87)	(−1.72)	(−1.69)	(−1.37)
Volatility	−0.020	−0.044	−0.016	−0.022	0.023	0.012	0.050	0.047
	(−0.15)	(−0.32)	(−0.11)	(−0.16)	(0.23)	(0.12)	(0.50)	(0.47)
Abnormal Earnings	0.020**	0.020**	0.020**	0.019**	0.009	0.009	0.008	0.008
	(2.20)	(2.11)	(2.18)	(1.99)	(1.34)	(1.27)	(1.19)	(1.05)
Tax Credit	−0.127	−0.124	−0.116	−0.068	−0.103	−0.103	−0.047	−0.027
	(−1.01)	(−0.97)	(−0.75)	(−0.43)	(−1.23)	(−1.21)	(−0.48)	(−0.27)
Loss Carry Forward	0.074	0.066	0.073	0.059	0.006	0.002	−0.001	−0.008
	(0.85)	(0.77)	(0.85)	(0.69)	(0.11)	(0.04)	(−0.01)	(−0.15)
Investment Grade	−0.069**	−0.063**	−0.069**	−0.058**	−0.065**	−0.062**	−0.061**	−0.055**
	(−4.11)	(−3.78)	(−3.91)	(−3.33)	(−4.96)	(−4.78)	(−4.29)	(−3.88)
CP Program	0.033**	0.031**	0.031**	0.025*	0.014**	0.013*	0.007	0.004
	(3.06)	(2.96)	(2.27)	(1.77)	(1.98)	(1.90)	(0.79)	(0.45)
CDS Count	0.007*		0.008*		0.004		0.005*	
	(1.90)		(1.80)		(1.52)		(1.79)	
CDS Bid-Ask		−0.175**		−0.194**		−0.082**		−0.107**
		(−2.77)		(−2.85)		(−2.27)		(−2.74)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X	X	X	X	X	X	X	X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.323	0.330	0.320	0.328	0.527	0.529	0.526	0.529

(*continued*)

Table 4
Continued

	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Maturity regression								
Book Leverage	−5.358*	−5.637**	−21.246*	−28.181**				
	(−1.91)	(−1.99)	(−1.94)	(−2.29)				
Market Leverage					−7.917**	−8.142**	−31.797**	−35.881**
					(−2.10)	(−2.15)	(−2.18)	(−2.37)
Industry Asset Maturity	0.364**	0.354**	0.384**	0.376**	0.358**	0.349**	0.364**	0.352**
	(2.18)	(2.11)	(2.32)	(2.28)	(2.15)	(2.09)	(2.22)	(2.14)
Market to Book	−0.046	0.048	−0.185	−0.037	−0.479	−0.406**	−1.928	−2.024**
	(−0.09)	(0.09)	(−0.35)	(−0.07)	(−0.89)	(−0.75)	(−1.96)	(−2.04)
Fixed Asset	0.151	0.376	1.954	3.032	0.133	0.337	1.913	2.463
	(0.06)	(0.14)	(0.69)	(1.05)	(0.05)	(0.13)	(0.71)	(0.91)
Profitability	1.399	1.023	1.971	1.588	0.456	0.067	−1.809	−2.709
	(0.28)	(0.21)	(0.39)	(0.31)	(0.09)	(0.01)	(−0.34)	(−0.51)
Size	0.126	0.205	−0.259	−0.293	0.166	0.246	−0.102	−0.038
	(0.35)	(0.57)	(−0.54)	(−0.61)	(0.47)	(0.70)	(−0.25)	(−0.09)
Volatility	8.828	7.979	10.313	9.343	9.365	8.579	12.492	11.783
	(1.09)	(1.01)	(1.25)	(1.14)	(1.17)	(1.09)	(1.46)	(1.39)
Abnormal Earnings	−0.199	−0.219	0.086	0.183	−0.239	−0.261	−0.067	−0.065
	(−0.39)	(−0.44)	(0.16)	(0.35)	(−0.47)	(−0.53)	(−0.13)	(−0.13)
Tax Credit	8.088	8.210	6.327	5.875	7.724	7.836	4.834	4.573
	(0.92)	(0.94)	(0.72)	(0.68)	(0.88)	(0.90)	(0.57)	(0.55)
Loss Carry Forward	−2.340	−2.611	−1.109	−1.047	−2.676	−2.953	−2.440	−2.786
	(−0.71)	(−0.81)	(−0.35)	(−0.34)	(−0.81)	(−0.90)	(−0.76)	(−0.89)
Investment Grade	0.580	0.824	−0.566	−0.648	0.459	0.697	−1.073	−0.991
	(0.65)	(0.94)	(−0.48)	(−0.54)	(0.51)	(0.78)	(−0.83)	(−0.77)
CP Program	−1.481**	−1.514**	−0.915	−0.745	−1.543**	−1.579**	−1.153*	−1.146*
	(−2.52)	(−2.57)	(−1.33)	(−1.08)	(−2.62)	(−2.67)	(−1.84)	(−1.85)
CDS Count	0.388*		0.504**		0.378*		0.465**	
	(1.74)		(2.17)		(1.70)		(2.06)	
CDS Bid-Ask		−5.754**		−10.102**		−5.410**		−7.941**
		(−2.08)		(−2.91)		(−1.98)		(−2.69)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X	X	X	X	X	X	X	X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.125	0.126	0.127	0.132	0.126	0.127	0.130	0.133

This table presents ordinary least squares (One Equation) and two-stage least squares (Two Equations) regression results of leverage and maturity on the explanatory variables in Table 3 plus CDS market liquidity proxies for the subsample of firms for which *CDS Trading* equals one. CDS Count is defined as the log number of daily CDS quotes on Bloomberg during year t (across all maturities). *Bid-Ask Spread* is defined as the average daily percentage bid-ask spread for CDSs on the firm's debt, as disseminated on Bloomberg, during year t, All other variables are defined in Tables 1 and 3. Time and industry fixed effects are included in all regressions. Standard errors are clustered at the firm level. * and ** denote significance levels of 10% and 5%, respectively. Constants are estimated but not reported. The sample is composed of nonfinancial firms in the S&P 500 index for which we observe a CDS contract trading during the 2002–2010 period, resulting in 1,578 firm/year observations.

Read:

- Restricting to firms that already have CDSs, more liquid CDS contracts are associated with more leverage and longer maturity.
- A one-standard-deviation increase in the number of CDS quotes raises book leverage by about **0.011** and market leverage by about **0.007**.
- A one-standard-deviation decrease in bid-ask spreads raises book leverage by about **0.021** and market leverage by about **0.012**.

- Debt maturity rises by roughly **8 to 13 months** when CDS liquidity improves.

Economic message. It is not only the existence of a CDS market that matters. How tradable the hedge is also matters, which strongly supports the hedging interpretation.

5.5 Table 5 and Appendix Table A5: IV results

Table 5
Instrumental variables approach

	Book Leverage				Market Leverage			
	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Leverage regression								
Debt Maturity	0.001	0.001	0.005	−0.011	0.000	0.000	0.001	−0.006
	(0.85)	(1.33)	(0.74)	(−1.01)	(0.38)	(0.97)	(0.28)	(−0.78)
Industry Book Leverage	0.306**	0.206**	0.302**	0.222**				
	(5.31)	(2.52)	(5.24)	(2.60)				
Industry Market Leverage					0.335**	0.239**	0.321**	0.264**
					(7.36)	(4.39)	(6.57)	(4.41)
Market to Book	−0.008	−0.013**	−0.007	−0.012**	−0.032**	−0.026**	−0.030**	−0.025**
	(−1.34)	(−2.57)	(−1.04)	(−2.33)	(−4.15)	(−3.54)	(−3.98)	(−3.44)
Fixed Asset	0.088**	0.028	0.066	0.049	0.043**	0.019	0.029	0.030
	(2.73)	(0.50)	(1.59)	(0.80)	(1.95)	(0.48)	(0.97)	(0.69)
Profitability	0.107	−0.055	0.123	−0.061	−0.089*	−0.161**	−0.063	−0.163**
	(1.40)	(−0.91)	(1.52)	(−1.04)	(−1.91)	(−4.52)	(−1.21)	(−4.55)
Size	−0.025**	−0.037**	−0.032**	−0.042**	−0.009**	−0.017*	−0.019**	−0.019*
	(−4.54)	(−2.71)	(−3.52)	(−2.84)	(−2.31)	(−1.66)	(−2.45)	(−1.81)
Volatility	−0.089	0.015	−0.109	−0.021	−0.042	−0.053	−0.081	−0.074
	(−0.76)	(0.10)	(−0.87)	(−0.14)	(−0.55)	(−0.60)	(−1.01)	(−0.81)
Abnormal Earnings	0.007	0.003	0.007	0.002	0.008	0.005	0.006	0.004
	(0.96)	(0.39)	(1.01)	(0.26)	(1.11)	(0.54)	(0.94)	(0.49)
CP Program	0.019**	0.006	0.028*	0.003	0.006	0.004	0.009	0.003
	(2.17)	(1.35)	(1.73)	(0.64)	(0.99)	(1.17)	(0.74)	(0.63)
Tax Credit	0.111	−0.077	0.056	0.060	0.030	−0.048	0.021	0.024
	(1.04)	(−0.50)	(0.39)	(0.31)	(0.39)	(−0.44)	(0.19)	(0.17)
Loss Carry Forward	0.034	0.023	0.040	−0.007	−0.009	−0.025	0.004	−0.041
	(0.69)	(0.53)	(0.77)	(−0.15)	(−0.27)	(−0.66)	(0.11)	(−0.98)
Rated	0.125**	0.145**	0.110**	0.165**	0.085**	0.057**	0.068**	0.068**
	(5.07)	(3.62)	(4.14)	(3.62)	(5.97)	(2.63)	(3.85)	(2.65)
Investment Grade	−0.093**	−0.029**	−0.097**	−0.025**	−0.081**	−0.030**	−0.085**	−0.028**
	(−7.09)	(−2.61)	(−7.33)	(−2.08)	(−7.57)	(−3.22)	(−7.46)	(−2.91)
CDS Trading IV	0.075**	0.097*	0.101**	0.184*	0.046**	0.099*	0.104**	0.146*
	(5.47)	(1.88)	(2.00)	(1.89)	(4.55)	(1.84)	(2.12)	(1.79)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X		X		X		X	
Firm Fixed Effects		X		X		X		X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.372	0.806	0.345	0.806	0.554	0.847	0.545	0.847

(*continued*)

Table 5
Continued

	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Maturity regression								
Book Leverage	1.008	2.730	−12.158	−2.483				
	(0.48)	(1.43)	(−1.01)	(−0.10)				
Market Leverage					−0.445	2.890	−18.165	10.459
					(−0.17)	(1.15)	(−1.34)	(0.52)
Industry Asset Maturity	0.246*	0.336**	0.285*	0.274	0.252*	0.339**	0.292*	0.278*
	(1.78)	(2.23)	(1.82)	(1.60)	(1.82)	(2.19)	(1.92)	(1.80)
Market to Book	−0.143	0.097	−0.256	0.028	−0.172	0.138	−0.720	0.337
	(−0.50)	(0.43)	(−0.86)	(0.08)	(−0.59)	(0.63)	(−1.45)	(0.61)
Fixed Asset	1.119	−0.291	2.249	0.360	1.240	−0.284	1.899	−0.148
	(0.55)	(−0.11)	(0.99)	(0.12)	(0.61)	(−0.11)	(0.92)	(−0.05)
Profitability	0.946	−0.227	2.992	−0.597	0.993	0.188	−0.008	1.429
	(0.28)	(−0.09)	(0.88)	(−0.20)	(0.29)	(0.07)	(−0.00)	(0.34)
Size	0.234	−0.175	−0.392	−0.378	0.204	−0.269	−0.370	−0.095
	(0.76)	(−0.24)	(−0.73)	(−0.27)	(0.70)	(−0.37)	(−0.84)	(−0.11)
Volatility	−0.286	−2.167	−2.472	−2.248	−0.342	−2.039	−2.148	−1.822
	(−0.05)	(−0.42)	(−0.44)	(−0.44)	(−0.07)	(−0.41)	(−0.39)	(−0.34)
Abnormal Earnings	−0.194	−0.089	−0.157	−0.078	−0.185	−0.097	−0.150	−0.123
	(−0.65)	(−0.39)	(−0.46)	(−0.34)	(−0.61)	(−0.42)	(−0.45)	(−0.53)
CP Program	−1.883**	−0.201	−1.593**	−0.182	−1.858**	−0.197	−1.704**	−0.238
	(−4.38)	(−0.77)	(−3.34)	(−0.66)	(−4.34)	(−0.76)	(−3.95)	(−0.91)
Tax Credit	10.159	10.086	12.247	10.335	10.282	10.181	11.168	10.408
	(1.45)	(1.27)	(1.60)	(1.29)	(1.48)	(1.28)	(1.58)	(1.28)
Loss Carry Forward	0.312	−1.991	1.024	−2.020	0.328	−1.953	0.530	−1.706
	(0.13)	(−0.85)	(0.41)	(−0.87)	(0.14)	(−0.84)	(0.22)	(−0.66)
Rated	0.858	1.352	2.107	2.149	1.032	1.571	1.937	1.121
	(0.86)	(0.99)	(1.16)	(0.54)	(1.04)	(1.16)	(1.30)	(0.66)
Investment Grade	0.680	0.393	−0.675	0.254	0.544	0.412	−1.075	0.633
	(1.00)	(0.54)	(−0.50)	(0.27)	(0.78)	(0.56)	(−0.78)	(0.73)
CDS Trading IV	2.480**	6.094*	5.299**	6.916	2.583**	6.369*	5.990**	5.278
	(3.07)	(1.72)	(2.32)	(1.14)	(3.30)	(1.77)	(2.60)	(1.00)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X		X		X		X	
Firm Fixed Effects		X		X		X		X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.143	0.680	0.129	0.680	0.142	0.680	0.131	0.680

Panels A and B present instrumental variables regressions for leverage and debt maturity. The instrument for *CDS Trading* is the average amount of foreign exchange derivatives used for hedging purposes (not trading) relative to total assets of the lead syndicate banks and bond underwriters that firms have done business with in the past five years (*Bank Foreign Exchange Derivatives*). We instrument the *CDS Trading* variable in all 8 of the regressions presented in this table. All variables are defined in Tables 1 and 3. The regression specifications shown in Columns 1, 3, 5, and 7 include time and industry fixed effects. Industry fixed effects and the *CDS Traded* dummy variable are replaced with firm fixed effects in the specifications shown in Columns 2, 4, 6, and 8. In the Two Equation specifications, the identifying variable for the leverage equation is *Industry Leverage*. The identifying variable for the debt maturity equation is *Industry Asset Maturity*. Results of the probit model estimation used to construct the instrumental variable are reported in Appendix Table A5. All standard errors are clustered at the firm level. * and ** denote significance levels of 10% and 5%, respectively. Constants are estimated but not reported. The sample consists of nonfinancial firms in the S&P 500 index for which lender and underwriter information is available (from DealScan and FISD, respectively). The bank holding companies of the lenders and underwriters must also be in the Federal Reserve's Call Report data during the 2002–2010 period, resulting in a total of 2,682 firm/year observations.

Read:

- Instrumenting CDS trading with *Bank Foreign Exchange Derivatives* leaves the qualitative results intact.
- The instrument has a strong first stage: incremental pseudo-$R^2 \approx 0.014$ and F-tests above 50.
- The IV estimates imply that a one-standard-deviation increase in instrumented CDS exposure raises leverage by about **0.020 to 0.055** and maturity by about **1 to 3 years**.

Economic message. The baseline results are unlikely to be driven solely by endogenous timing of CDS creation. Even when CDS status is instrumented, the capital-structure effect remains large.

Table 6
Changes in leverage and debt maturity around CDS introduction—CDS firms versus matched sample

	CDS Firms		Control Firms		CDS – Control	
	Year (End) t − 1 to t	Year (End) t − 1 to t+1	Year (End) t − 1 to t	Year (End) t − 1 to t+1	Year (End) t − 1 to t	Year (End) t − 1 to t+1
Δ Book Leverage	0.014**	0.006	−0.011**	−0.018**	0.025**	0.025**
	(2.40)	(0.89)	(−2.73)	(−2.67)	(3.84)	(2.58)
Δ Market Leverage	0.012**	0.003	−0.012**	−0.015**	0.024**	0.019**
	(2.11)	(0.50)	(−2.28)	(−2.15)	(3.55)	(2.15)
Δ Debt Maturity	0.146	−0.054	−0.974**	−0.955**	1.120**	0.901
	(0.52)	(−0.16)	(−2.29)	(−2.11)	(2.07)	(1.49)
Δ Bond to Total Asset	0.006	0.019**	−0.006	0.002	0.013	0.016
	(0.94)	(2.18)	(−1.20)	(0.33)	(1.55)	(1.59)
Δ Bond Maturity	−0.146	−0.347	−0.665	−0.691	0.517	0.368
	(−0.54)	(−0.96)	(−1.40)	(−1.34)	(0.90)	(0.56)
Δ Loan to Total Asset	−0.001	−0.002	0.002	−0.008**	−0.004	0.005
	(−0.42)	(−0.60)	(0.58)	(−2.22)	(−0.75)	(0.86)
Δ Loan Maturity	−0.129	−0.172	0.018	0.060	−0.148	−0.290
	(−0.52)	(−0.71)	(0.12)	(0.28)	(−0.50)	(−0.87)
Δ Debt / Debt	0.212**	0.253**	−0.012	0.081	0.224**	0.172**
	(4.08)	(4.53)	(−0.61)	(1.44)	(4.13)	(2.11)
Δ Bond / Bond	0.135**	0.254**	−0.010	0.135**	0.145**	0.119
	(2.99)	(4.61)	(−0.41)	(2.47)	(2.83)	(1.53)
Δ Public Bond / Public Bond	0.106**	0.189**	0.025	0.135**	0.082**	0.054
	(3.00)	(4.15)	(1.42)	(2.72)	(2.02)	(0.78)
Δ Private Bond / Private Bond	0.000	0.009	−0.042**	−0.045**	0.042**	0.055**
	(0.02)	(0.48)	(−2.93)	(−2.42)	(2.32)	(2.08)
Δ Loan / Loan	0.021	0.019	0.053**	0.004	−0.031	0.016
	(1.04)	(0.84)	(2.28)	(0.18)	(−1.01)	(0.48)
Δ Other / Other	0.056*	−0.021	−0.027	−0.029	0.082**	0.009
	(1.77)	(−0.89)	(−1.42)	(−1.41)	(2.06)	(0.27)

This table presents univariate analysis of changes in leverage and maturity during the year of CDS introduction (end of year $t-1$ to end of year t) and in the year following CDS introduction (end of year $t-1$ through the end of $t+1$). All variables are expressed in changes relative to a matched firm. The matched sample of non-CDS firms is chosen based on propensity scores obtained by estimating a probit model of the likelihood of CDS trading. The probit estimation results are in the Appendix Table A6. In such a model, to guarantee that no outcome variable is included as a regressor, all independent variables are lagged by one year. Further, to guarantee that firms are on parallel trends prior to CDS introduction, one-year changes in leverage and debt maturity are also included as regressors. Variable definitions for the changes in leverage, debt maturity, and the bond and loan components of leverage (i.e., those in the top portion of the table) are given in Table 1. We also include percentage changes in debt and various debt components. Δ *Debt / Debt* is the percentage change in total debt. Δ *Bond / Bond* is the percentage change in total bonds outstanding, as reported in Capital IQ. Δ *Public Bond / Public Bond* is the percentage change in public bonds outstanding, from the FISD data. Δ *Private Bond / Private Bond* is the percentage change in private bonds outstanding, where the amount of private bonds outstanding is calculated as the difference between total bonds outstanding (from Capital IQ) and total public bonds (from FISD). Δ *Loan / Loan* and Δ *Other / Other* are the percentage changes in total loans and other debt outstanding, respectively, as reported in Capital IQ. We report the difference-in-difference estimates for the CDS firms and the closest control firm based on propensity score. * and ** denote significance levels of 10% and 5%, respectively. There are 122 matches. The sample is composed of nonfinancial firms in the S&P 500 index during the 2002–2010 period.

5.6 Table 6 and Appendix Table A6: matched-sample CDS introduction

Read:

- Around the year of CDS introduction, book leverage rises by about **0.014** for treated firms and falls by about **0.011** for matched controls, yielding a difference-in-differences of about **0.025**.
- Market leverage shows a similar treatment-control difference of about **0.024 to 0.025**.
- Debt maturity shows a treatment-control difference of about **one year** in the year of introduction.
- Total debt rises sharply for treated firms relative to controls, with the increase especially concentrated in public bonds.

Economic message. The CDS event itself is followed by a clear balance-sheet change: firms do not just look different in levels; they move differently right around adoption.

5.7 Figures 1 and 2: when CDS matters most over time

Read:

- Figure 1 plots year-by-year CDS coefficients for leverage and maturity. The coefficients are

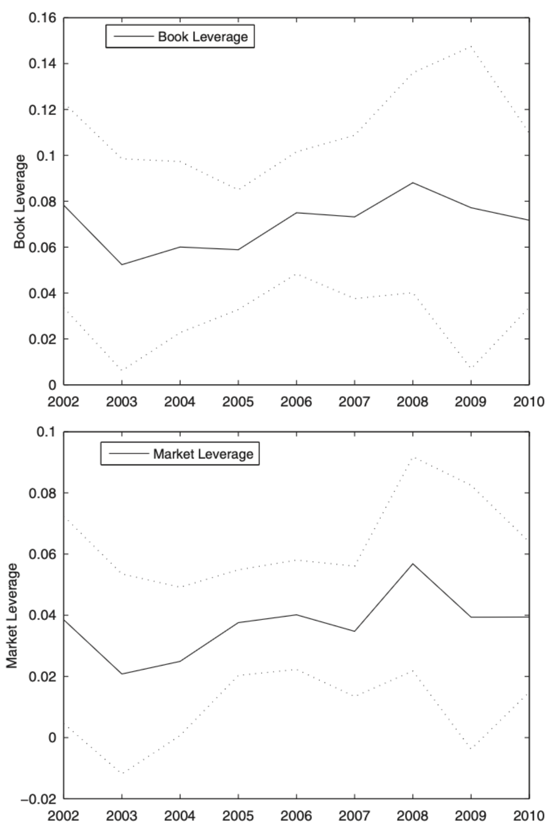

Figure 1
Impact of *CDS Trading*, by year
These figures show the time series of estimated coefficients on the *CDS Trading* dummy variable in the *Book Leverage*, *Market Leverage* and *Debt Maturity* regressions. We estimate parameters of the 2SLS specifications reported in Columns (3) and (7) of Panel A of Table 3 for leverage and of Panel (B) of Table 3 for maturity, obtaining four time-series of coefficients. The "Book Leverage" lines show the time series of estimated coefficients from the regressions using book leverage. The "Market Leverage" lines show the coefficients from the regressions using market leverage. The dotted lines indicate 5% and 95% confidence bands around the series of point estimates.

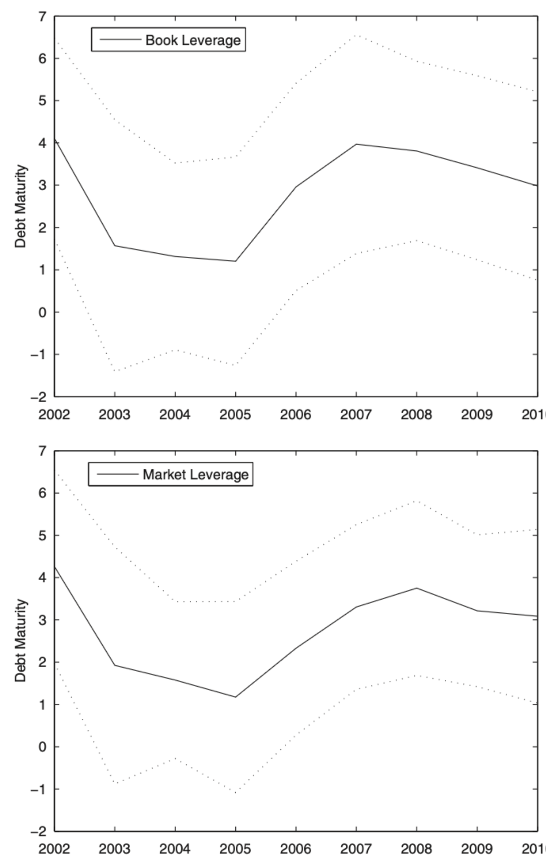

Figure 1
Continued

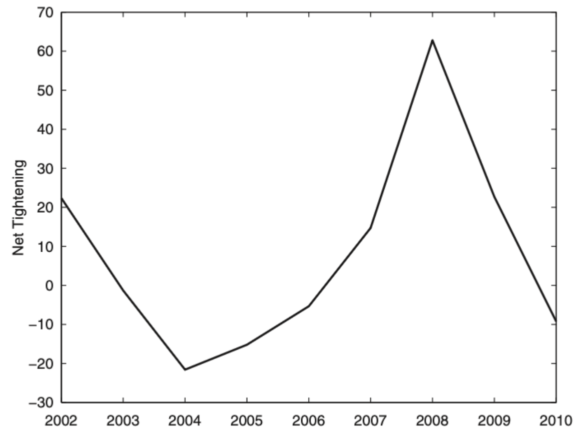

Figure 2
Net tightening of credit supply, by year
This figure shows the net percentage of senior loan officers reporting commercial and industrial loan tightening to large- and medium-sized firms over the 2002–2010 sample period. The net percentage of loan officers is the number of respondents reporting that they have tightened standards on commercial and industrial loans minus the number that have loosened standards, divided by the total number of survey respondents. Higher values indicate tightening of credit supply. Data are from the Federal Reserve's Senior Loan Officer Opinion Survey.

positive throughout and display a pronounced U-shape, peaking when credit conditions are tighter.

- Figure 2 plots the net share of senior loan officers tightening standards for commercial and industrial loans. It tracks the CDS-effect series closely.
- The correlations between the loan-officer tightening series and the CDS coefficients are about **0.79** and **0.72** for leverage, and **0.73** and **0.76** for debt maturity.

Economic message. CDSs matter most when credit supply matters most. That time-series pattern is hard to square with a pure firm-demand explanation.

5.8 Tables 7 and 8: local defaults and bank writedowns

Read:

- Table 7 uses within-state defaults outside the firm's industry as a local supply shock. The interaction *State Default* × *CDS Trading* is positive in most specifications.
- A one-standard-deviation increase in same-state default intensity allows CDS firms to maintain book leverage roughly **0.47 percentage points** higher and market leverage roughly **0.35 percentage points** higher than non-CDS firms facing the same shock.
- The same shock lets CDS firms keep debt maturity about **seven months** longer than comparable non-CDS firms.
- Table 8 uses crisis-period bank writedowns. The interaction with CDS trading is again positive: CDS firms are less exposed when related lenders and underwriters are in distress.

Economic message. These are the paper's cleanest supply-side results. When credit gets scarce locally or at connected banks, CDS firms are relatively shielded.

Table 7
Regional supply shocks: Within state debt defaults

	Book Leverage				Market Leverage			
	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Leverage regression								
Debt Maturity	0.001	0.000	0.005	−0.003	0.000	−0.000	0.002	−0.003
	(1.42)	(0.60)	(0.72)	(−0.49)	(0.75)	(−0.08)	(0.39)	(−0.70)
Industry Book Leverage	0.366**	0.199**	0.371**	0.207**				
	(6.48)	(2.92)	(6.57)	(2.91)				
Industry Market Leverage					0.375**	0.275**	0.379**	0.283**
					(8.55)	(5.39)	(7.91)	(5.47)
Market to Book	−0.006	−0.013**	−0.006	−0.013**	−0.030**	−0.026**	−0.030**	−0.026**
	(−0.78)	(−2.92)	(−0.76)	(−2.93)	(−5.41)	(−4.11)	(−5.38)	(−4.12)
Fixed Asset	0.095**	0.049	0.081*	0.049	0.055**	0.004	0.049*	0.003
	(2.93)	(0.94)	(1.91)	(0.95)	(2.57)	(0.09)	(1.68)	(0.08)
Profitability	0.128*	−0.077	0.137*	−0.079	−0.075*	−0.167**	−0.071	−0.169**
	(1.78)	(−1.54)	(1.90)	(−1.61)	(−1.74)	(−4.50)	(−1.61)	(−4.55)
Size	−0.023**	−0.023**	−0.023**	−0.023**	−0.006	−0.010	−0.006	−0.010
	(−4.39)	(−2.02)	(−4.45)	(−2.07)	(−1.59)	(−1.24)	(−1.64)	(−1.29)
Volatility	−0.060	−0.040	−0.039	−0.055	−0.052	−0.074	−0.044	−0.087
	(−0.55)	(−0.35)	(−0.35)	(−0.47)	(−0.79)	(−1.06)	(−0.66)	(−1.23)
Abnormal Earnings	0.007	0.000	0.007	0.000	0.005	0.002	0.005	0.002
	(1.09)	(0.03)	(1.13)	(0.05)	(0.95)	(0.32)	(0.97)	(0.34)
Tax Credit	0.078	−0.109	0.017	−0.086	0.005	−0.069	−0.021	−0.048
	(0.72)	(−0.71)	(0.11)	(−0.53)	(0.07)	(−0.65)	(−0.18)	(−0.44)
Loss Carry Forward	0.027	0.018	0.024	0.015	0.011	−0.002	0.010	−0.004
	(1.40)	(0.54)	(1.21)	(0.47)	(1.09)	(−0.07)	(0.89)	(−0.16)
Rated	0.110**	0.137**	0.111**	0.140**	0.081**	0.067**	0.081**	0.069**
	(5.23)	(3.71)	(5.23)	(3.73)	(6.25)	(3.75)	(6.23)	(3.77)
Investment Grade	−0.089**	−0.032**	−0.093**	−0.032**	−0.080**	−0.034**	−0.082**	−0.034**
	(−7.09)	(−2.96)	(−6.68)	(−2.94)	(−8.44)	(−3.81)	(−7.31)	(−3.78)
CP Program	0.019**	0.005	0.026*	0.005	0.004	0.003	0.007	0.003
	(2.18)	(1.17)	(1.70)	(1.14)	(0.73)	(0.85)	(0.67)	(0.80)
CDS Traded	0.005		0.002		0.001		−0.000	
	(0.41)		(0.15)		(0.16)		(−0.01)	
CDS Trading	0.052**	0.015**	0.047**	0.017**	0.031**	0.008	0.029**	0.009
	(5.01)	(2.07)	(3.49)	(2.01)	(4.51)	(1.40)	(3.14)	(1.55)
State Defaults	−0.141*	−0.087	−0.157*	−0.087	−0.057	−0.010	−0.063	−0.008
	(−1.73)	(−1.51)	(−1.89)	(−1.52)	(−1.01)	(−0.22)	(−1.09)	(−0.20)
State Defaults × CDS Trading	0.229*	0.234**	0.241*	0.237**	0.145*	0.131*	0.148*	0.131*
	(1.82)	(2.50)	(1.94)	(2.54)	(1.68)	(1.81)	(1.70)	(1.81)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X		X		X		X	
Firm Fixed Effects		X		X		X		X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.341	0.794	0.339	0.794	0.537	0.837	0.537	0.838

Table 7
Continued

	One Equation		Two Equations		One Equation		Two Equations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Maturity regression								
Book Leverage	1.964	1.078	−10.347	−2.254				
	(1.02)	(0.65)	(−1.17)	(−0.10)				
Market Leverage					0.163	0.053	−16.297	2.796
					(0.07)	(0.02)	(−1.52)	(0.19)
Industry Asset Maturity	0.235*	0.483**	0.293**	0.485**	0.244*	0.484**	0.308**	0.482**
	(1.77)	(2.72)	(2.01)	(2.75)	(1.83)	(2.71)	(2.13)	(2.74)
Market to Book	−0.005	−0.036	−0.144	−0.081	−0.022	−0.049	−0.587	0.026
	(−0.02)	(−0.15)	(−0.55)	(−0.22)	(−0.08)	(−0.20)	(−1.41)	(0.06)
Fixed Asset	1.239	−1.921	2.612	−1.729	1.449	−1.856	2.531	−1.886
	(0.67)	(−0.84)	(1.27)	(−0.65)	(0.78)	(−0.81)	(1.30)	(−0.82)
Profitability	−2.174	−0.522	−0.494	−0.793	−1.894	−0.600	−3.317	−0.098
	(−0.67)	(−0.22)	(−0.15)	(−0.26)	(−0.59)	(−0.24)	(−0.98)	(−0.03)
Size	0.102	−0.399	−0.177	−0.473	0.058	−0.423	−0.021	−0.401
	(0.36)	(−0.76)	(−0.49)	(−0.63)	(0.21)	(−0.81)	(−0.08)	(−0.74)
Volatility	−4.671	−5.065	−5.168	−5.178	−4.752	−5.099	−5.018	−4.935
	(−1.08)	(−1.16)	(−1.21)	(−1.18)	(−1.10)	(−1.18)	(−1.18)	(−1.09)
Abnormal Earnings	−0.050	0.028	0.015	0.027	−0.040	0.027	0.018	0.023
	(−0.24)	(0.18)	(0.07)	(0.18)	(−0.20)	(0.18)	(0.08)	(0.15)
Tax Credit	12.275*	6.595	13.716**	6.313	12.502*	6.502	12.534*	6.626
	(1.85)	(0.81)	(2.00)	(0.79)	(1.90)	(0.80)	(1.93)	(0.83)
Loss Carry Forward	0.918	−0.702	1.142	−0.640	0.953	−0.682	1.002	−0.680
	(1.14)	(−0.57)	(1.38)	(−0.50)	(1.20)	(−0.56)	(1.23)	(−0.56)
Rated	−0.405	0.798	1.075	1.270	−0.184	0.946	1.271	0.751
	(−0.43)	(0.74)	(0.80)	(0.38)	(−0.20)	(0.88)	(1.00)	(0.53)
Investment Grade	1.163*	0.129	−0.002	0.019	0.992	0.095	−0.418	0.192
	(1.90)	(0.20)	(−0.00)	(0.02)	(1.58)	(0.14)	(−0.38)	(0.24)
CP Program	−1.837**	0.020	−1.588**	0.038	−1.798**	0.026	−1.697**	0.018
	(−4.23)	(0.07)	(−3.48)	(0.13)	(−4.17)	(0.09)	(−3.94)	(0.06)
CDS Traded	0.959		1.087		0.978		1.053	
	(1.37)		(1.53)		(1.40)		(1.51)	
CDS Trading	0.884	0.550	1.566**	0.608	0.983*	0.566	1.533**	0.541
	(1.53)	(1.44)	(2.09)	(1.00)	(1.71)	(1.49)	(2.29)	(1.28)
State Defaults	−4.321	−4.409	−5.364	−4.454	−4.648	−4.500	−4.712	−4.648
	(−0.90)	(−1.60)	(−1.14)	(−1.58)	(−0.96)	(−1.63)	(−0.99)	(−1.60)
State Defaults × CDS Trading	17.090**	10.554**	17.670**	10.760**	17.557**	10.787**	17.510**	10.831**
	(1.96)	(2.13)	(2.04)	(2.18)	(2.02)	(2.17)	(2.01)	(2.19)
Time Fixed Effects	X	X	X	X	X	X	X	X
Industry Fixed Effects	X		X		X		X	
Firm Fixed Effects		X		X		X		X
Clustered SE	X	X	X	X	X	X	X	X
Adj-R^2	0.124	0.678	0.125	0.678	0.123	0.678	0.126	0.678

This table presents ordinary least squares (One Equation) and two-stage least squares (Two Equations) regression results of leverage and debt maturity on the explanatory variables in Table 3, plus a local credit market supply shock variable. *State Default* is the credit market supply shock variable, defined as defined as the dollar value of all defaulted debt by Compustat firms outside of the sample firm's industry and headquartered in the sample firm's state during year $t-1$, divided by the total book value of debt of all Compustat firms outside of the sample firm's industry and headquartered in the state at the beginning of year $t-1$. *State Default × CDS Trading* is the credit supply shock and *CDS Trading* interaction variable. All other variables are defined in Tables 1 and 3. Leverage regression results are shown in Panel A. Results of the Maturity regression are in Panel B. The identifying variable for the leverage equation is *Industry Leverage*. The identifying variable for the debt maturity equation is *Industry Asset Maturity*. The regression specifications shown in Columns 1, 3, 5, and 7 include time and industry fixed effects. Industry fixed effects and the *CDS Traded* dummy variable are replaced with firm fixed effects in the specifications shown in Columns 2, 4, 6, and 8. Standard errors are clustered at the firm level. * and ** denote significance levels of 10% and 5%, respectively. Constants are estimated but not reported. The sample is composed of nonfinancial firms in the S&P 500 index during the 2002–2010 period, granting a total of 3,168 firm/year observations.

Table 8
Bank specific supply shocks: Writedowns during the financial crisis

	Book Leverage		Market Leverage	
	One Equation (1)	Two Equations (2)	One Equation (3)	Two Equations (4)
Panel A: Leverage regression				
Debt Maturity	0.001	0.005	0.000	0.001
	(0.77)	(0.84)	(0.10)	(0.38)
Industry Book Leverage	0.281**	0.288**		
	(4.17)	(4.23)		
Industry Book Leverage			0.252**	0.256**
			(5.02)	(4.91)
Market to Book	−0.001	−0.001	−0.041**	−0.041**
	(−0.11)	(−0.12)	(−6.23)	(−6.23)
Fixed Asset	0.095**	0.081*	0.056**	0.051*
	(2.65)	(1.86)	(2.20)	(1.73)
Profitability	−0.013	−0.001	−0.128**	−0.123**
	(−0.15)	(−0.01)	(−2.17)	(−2.09)
Size	−0.028**	−0.028**	−0.011**	−0.011**
	(−4.47)	(−4.54)	(−2.49)	(−2.54)
Volatility	−0.036	−0.018	0.069	0.074
	(−0.26)	(−0.13)	(0.72)	(0.78)
Abnormal Earnings	0.003	0.002	0.000	0.000
	(0.23)	(0.15)	(0.03)	(0.00)
Tax Credit	0.092	0.036	0.018	0.001
	(0.71)	(0.25)	(0.20)	(0.01)
Loss Carry Forward	0.008	0.006	−0.025	−0.025
	(0.13)	(0.09)	(−0.62)	(−0.63)
Rated	0.130**	0.133**	0.094**	0.095**
	(4.71)	(4.81)	(5.32)	(5.31)
Investment Grade	−0.088**	−0.096**	−0.081**	−0.084**
	(−4.88)	(−4.58)	(−5.71)	(−5.07)
CP Program	0.026**	0.033**	0.012	0.015
	(2.09)	(2.00)	(1.56)	(1.38)
CDS Trading	0.048**	0.040**	0.026**	0.024*
	(3.24)	(2.25)	(2.62)	(1.93)
Bank Writedowns	−0.006	−0.006	−0.005*	−0.005*
	(−1.61)	(−1.54)	(−1.96)	(−1.91)
Bank Writedowns × CDS Trading	0.006	0.005	0.006**	0.005*
	(1.58)	(1.07)	(2.08)	(1.72)
Time Fixed Effects	X	X	X	X
Industry Fixed Effects	X	X	X	X
Firm Fixed Effects				
Clustered SE	X	X	X	X
Adj-R^2	0.362	0.362	0.548	0.548

(*continued*)

Table 8
Continued

	One Equation (1)	Two Equations (2)	One Equation (3)	Two Equations (4)
Panel B: Debt maturity regression				
Book Leverage	0.767	−16.846		
	(0.37)	(−1.29)		
Market Leverage			−1.038	−27.515*
			(−0.39)	(−1.67)
Industry Asset Maturity	0.294**	0.360**	0.299**	0.364**
	(2.26)	(2.62)	(2.31)	(2.72)
Market to Book	0.006	−0.064	−0.042	−1.183
	(0.02)	(−0.21)	(−0.13)	(−1.58)
Fixed Asset	−0.087	1.797	0.062	1.751
	(−0.04)	(0.71)	(0.03)	(0.77)
Profitability	−1.684	−2.504	−1.897	−6.419
	(−0.52)	(−0.74)	(−0.58)	(−1.39)
Size	0.067	−0.435	0.033	−0.274
	(0.28)	(−0.96)	(0.14)	(−0.89)
Volatility	−4.250	−3.051	−4.021	0.502
	(−0.89)	(−0.58)	(−0.84)	(0.08)
Abnormal Earnings	0.158	0.239	0.163	0.199
	(0.34)	(0.52)	(0.35)	(0.44)
Tax Credit	8.004	9.722	8.073	7.932
	(1.13)	(1.29)	(1.15)	(1.15)
Loss Carry Forward	0.537	0.532	0.501	−0.418
	(0.17)	(0.17)	(0.16)	(−0.14)
Rated	−0.615	1.916	−0.398	2.325
	(−0.49)	(0.95)	(−0.32)	(1.24)
Investment Grade	1.858**	0.204	1.697**	−0.576
	(3.04)	(0.16)	(2.79)	(−0.39)
CP Program	−1.581**	−1.079*	−1.544**	−1.151**
	(−3.08)	(−1.66)	(−3.00)	(−1.99)
CDS Trading	1.790**	2.687**	1.858**	2.587**
	(2.96)	(3.05)	(3.10)	(3.53)
Bank Writedowns	−0.060	−0.200	−0.073	−0.237
	(−0.31)	(−0.90)	(−0.37)	(−1.08)
Bank Writedowns × CDS Trading	0.325*	0.460**	0.337*	0.502**
	(1.71)	(2.13)	(1.77)	(2.31)
Time Fixed Effects	X	X	X	X
Industry Fixed Effects	X	X	X	X
Firm Fixed Effects				
Clustered SE	X	X	X	X
Adj-R^2	0.170	0.174	0.170	0.176

This table presents ordinary least squares (One Equation) and two-stage least squares (Two Equations) regression results of leverage and debt maturity on the explanatory variables in Table 3 plus a financial crisis period bank writedown variable. *Bank Writedown* is the credit market supply shock variable, defined as the value of writedown losses from structured finance products to the book value of the equity of the banks that have served either as underwriter or lead loan syndicate member for firm i during the years $t-1$ through $t-5$. *Bank Writedown × CDS Trading* is the credit supply shock and *CDS Trading* interaction variable. All other variables are defined in Tables 1 and 3. Leverage regression results are shown in Panel A. Results of the Maturity regression are in Panel B. The identifying variable for the leverage equation is *Industry Leverage*. The identifying variable for the debt maturity equation is *Industry Asset Maturity*. All regression specifications include time and industry fixed effects. Standard errors are clustered at the firm level. * and ** denote significance levels of 10% and 5%, respectively. Constants are estimated but not reported. The sample is composed of nonfinancial firms in the S&P 500 index during the recent financial crisis period (2008–2010), granting a total of 1,237 firm/year observations.

6 Short summary

- The paper studies whether CDS markets change firm leverage and debt maturity by easing the supply of debt capital.
- Using nonfinancial S&P 500 firms from 2002 to 2010, it finds that firms with traded CDSs on their debt carry more leverage and issue debt at longer maturities.
- The baseline result survives firm fixed effects, within-CDS liquidity tests, an IV strategy, and a matched-sample event study.
- The effect works mainly through bonds rather than loans.
- The impact of CDSs becomes stronger when aggregate, local, or lender-specific credit constraints tighten.
- The paper's main message is that CDSs relax supply frictions in corporate debt markets.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the availability of CDS contracts on a firm's debt is associated with economically meaningful increases in both leverage and debt maturity. The evidence is strongest when interpreted as a supply-side hedging effect, not as a simple reflection of firm demand for debt.

7.2 Economic intuition

A lender or bond investor is willing to supply more credit when default risk can be insured. That relaxed constraint can show up as more debt, longer maturity, or both. The paper argues that CDS markets do exactly that, especially when credit markets are stressed and hedging capacity becomes scarce.

7.3 Takeaway

A useful way to remember the paper is: *CDSs expand debt capacity by making risky credit easier to hold.* In this setting, the effect is not just a spread effect. It changes how much firms borrow, how long they can borrow for, and how resilient that financing is when credit conditions deteriorate.